THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In Euclid's subtraction algorithm, the pair (m, n) with $m \geq n$ is replaced by: 1

(a) $(m,\ n+1)$ **(b)** $(n,\ m \times n)$

(c) the numbers n and $m - n$ **(d)** $(m+n,\ n)$

2. Using Euclid's subtraction algorithm for gcd(88, 66), the number of subtraction steps needed is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

3. Set x = 4. Repeat 3 times: x = 2x − 2. The final value of x is: 1

(a) 20 **(b)** 19 **(c)** 18 **(d)** 17

4. By Euclid's algorithm, gcd(184, 168) is: 1

(a) 7 **(b)** 10 **(c)** 9 **(d)** 8

5. Adding 8208 and 3739 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

6. The divisor algorithm applied to 36 produces a list with how many entries? 1

(a) 10 **(b)** 11 **(c)** 9 **(d)** 8

7. By Euclid's algorithm, gcd(114, 34) is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

8. The value of $233 \bmod 12$ (remainder on dividing 233 by 12) is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

9. Every algorithm has input, output and: 1

(a) an unlimited number of steps **(b)** no stopping rule

(c) a finite sequence of well-defined steps **(d)** random choices

10. Using Euclid's subtraction algorithm for gcd(81, 63), the number of subtraction steps needed is: 1

(a) 6 **(b)** 5 **(c)** 8 **(d)** 7

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Run the divisor algorithm for 31: check j = 1, 2, …, 31. Write the list of divisors and say how many numbers are checked. 2

12. Apply Euclid's subtraction algorithm to find gcd(63, 39). Write the sequence of pairs and count the subtractions. 2

13. Apply Euclid's subtraction algorithm to find gcd(58, 40). Write the sequence of pairs and count the subtractions. 2

14. Run the divisor algorithm for 52: check j = 1, 2, …, 52. Write the list of divisors and say how many numbers are checked. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply Euclid's subtraction algorithm to find gcd(116, 68). Write the sequence of pairs and count the subtractions. 3

16. Apply Euclid's subtraction algorithm to find gcd(180, 100). Write the sequence of pairs and count the subtractions. 3

17. Add 7950 and 2826 using the column algorithm. State in which columns a carry is produced. 3

OR

Apply Euclid's subtraction algorithm to find gcd(102, 60). Write the sequence of pairs and count the subtractions. 3

18. Apply Euclid's subtraction algorithm to find gcd(189, 153). Write the sequence of pairs and count the subtractions. 3

OR

Add 6953 and 5580 using the column algorithm. State in which columns a carry is produced. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace: x = 9; repeat 3 times: x = 2x − 4. Write the value of x after each repetition and the final value. 5

OR

Trace: s = 0; while n > 0, add (n mod 10) to s and replace n by its quotient on division by 10. Start with n = 5092. Write the digits picked and the final s. 5

20. Trace: x = 8; repeat 4 times: x = 2x − 4. Write the value of x after each repetition and the final value. 5

OR

Trace: s = 0; while n > 0, add (n mod 10) to s and replace n by its quotient on division by 10. Start with n = 2990. Write the digits picked and the final s. 5

END OF QUESTION PAPER — Total: 40 Marks